IT'IS FOUNDATION	Project	Document Name	Rev.
	0xxx	**CRR-xxx**	**0**

Report Front Page Title

	Name	**Function**	**Date**	**Signature**
Author	Reviewing Author	Project Leader, IT'IS Foundation		
Approval	Niels Kuster	Quality Manager, IT'IS Foundation		

Prof. Dr. Niels Kuster

Director, IT'IS Foundation
Professor, Swiss Federal Institute of Technology – ETH Zurich
Phone +41 44 245 9690 · kuster@itis.swiss

Report Front Page Title

conducted by

Front Page Authors

Zurich, DD Month 2021

Revision History

Revision 1

- Improvement of the post-processing of the radiated test measurements in high and low permittivity medium, using a newer version of the “touchless” code which increases fidelity for devices with many small electrodes. The revised deposited power estimates are uniformly lower than the original evaluations, validating the conservativeness of the mixed-medium models, which are derived on the basis of the original evaluations. The test reports in Annex Y have been revised accordingly.
- Incorporation of measurements of additional samples. Taking maximum deposited power for each electrode type and frequency across all measurements resulted in an increase of the scaling factor of the mixed-medium model, and corresponding Tier 4- estimates of deposited power and temperature rise, of the implant at 128 MHz, compared to Revision 0. Tables X and Annex Y have been revised accordingly, and the corresponding measurements added to Annex X.

Executive Summary

The executive summary should be a brief overview of the objectives, methods, results, and conclusions (including limitations) of the work. Use SI units, like so: 50 cm.

Figure 1: Numerically derived deposited power factor as a function of wire break length from the tip electrode. Approximate half-wavelength in tissue simulating medium at 64 MHz is indicated.

Wire	Electrode							
break	1	2	3	4	5	6	7	8
1	–0.4	–3.3	–3.9	–2.6	–3.6	–3.9	–3.9	–3.8
2	–2.7	–2.9	–3.3	–2.5	–3.7	–3.1	–4.4	–3.1
3	–2.6	–2.3	–12.0	–2.8	–3.1	–3.8	–4.4	–3.2
4	–2.6	–2.2	–3.4	–14.5	–3.9	–3.3	–3.6	–3.9
5	–3.1	–2.3	–3.3	–2.6	–9.4	–4.0	–4.3	–3.7
6	–2.7	–2.8	–3.5	–2.6	–3.6	–2.3	–4.2	–3.8
7	–3.3	–3.5	–3.5	–2.8	–3.9	–3.8	–4.5	–4.0
8	–2.7	–2.9	–3.7	–2.6	–3.7	–3.8	–4.1	–5.9

Table 1: Estimated difference, in dB, in Tier 3 95% deposited power at maximum landmark around each electrode for a 50 cm lead with the indicated wire broken at 29.5 cm, for ViP model Duke over all clinical routings and exposure conditions in CRR-0xxx, compared to the result of the unbroken lead tip (590 mW), using numerically derived transfer functions. The numerically predicted 95% deposited power in Duke of electrode 1 for a break at wire 1 at the resonant length was 532 mW, whereas the validation measurements of that case gave a 95% deposited power in Duke of up to 331 mW.

The study was performed by the IT'IS laboratory in Zurich. The dosimetric scanner employed was the DASY6 from Schmid & Partner Engineering AG, Zurich. All equipment was appropriately calibrated, and the procedures used were in accordance with all the standards mentioned above.

In summary, the maximum SAR value of the device for the tested configurations is:

Phantom	f /(MHz)	Separation /(mm)	1 g psa. SAR ** /(W/kg)	10 g psa. SAR ** /(W/kg)
ELI4 Phantom	6.78	0	3.79 ± 23.7% *	2.54 ± 23.5% *

* Worst-case uncertainty of the DASY6 system with the the ELI4 phantom. ($k = 2$)

** Due to differences in measured EUT output voltage compared to the value measured by Customer Name and large power drifts, the results have been extrapolated to represent the worst-case exposure estimation. Non-extrapolated values are smaller: the SAR extrapolation factor to account for the output voltage difference was 1.7, and the factor to account for the worst-case power drift was 1.4; therefore, a total extrapolation factor of 2.4 was applied to the above results.

Contents

1 Objectives

Background and objectives of the project.

The following tasks were performed:

- item
- next item (can also link to sections if apppropriate

The EUT, a wireless power transfer system operating at X frequency, was tested based on the requirements defined by IEC/IEEE 62209-1528-2020. Dosimetric compliance was tested with the recommendations for SAR exposure in the head and trunk for the general public according to:

- ICNIRP "Guidelines for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to 300 GHz)" [1]
- Directive of the European Commission "On the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz)" [2]
- US "47 CFR § 1.1310 - Radiofrequency radiation exposure limits" [3]

The EUT was tested for compliance with localized 1 g, and 10 g averaged SAR in the head and trunk. Averaged SAR limits for general public exposure of [1–3] have the following maximum permissible level:

SAR avg. mass	SAR limit
1 g	1.6 W/kg
10 g	2.0 W/kg

2 Measurement System

Software Tools	Sim4Life	6.0
	IMAnalytics	3.0
	ViP	V3.1 (Fats V3.2)
	MRIxViP	2.1

Table 2: Software tool versions used in the numerical transfer function generation and Tier 3 evaluation. The verified FDTD solver of Sim4Life was used to derive all the numerical results.

DASY	Type:	DASY52 NEO
	Software version:	52.10.4.1535
	Postprocessor version:	14.6.14.7501
	Manufacturer:	Schmid & Partner Engineering AG (CH)
DAKS-12	Type:	DAKS-12
	Probe:	S/N 1000
	Probe manufacturer:	Schmid & Partner Engineering AG (CH)
	VNA/Reflectometer:	R60
	VNA Serial No:	170717
	VNA manufacturer:	Copper Mountain (US)
	Last calibrated:	10 June 2020
	Next calibration:	June 2021
	Software version:	DAK v2.6.1.7
Conductivity Meter	Type:	GMH 5430
	Serial No:	3440071
	Last calibrated:	25 September 2020
	Next calibration:	September 2021
	Manufacturer:	Greisinger GmbH (DE)
piX system	Type:	piX at 64 MHz
	piX analyzer:	S/N 1001
	piX excitor:	piXE64HPV1 ITIS-0062
	Phantom:	piXPhantomV3
	Manufacturer:	ZMT Zurich MedTech AG (CH)
TDS E-Field System	Type:	E1TDSz/MRI
	S/N Probe + RU:	1054 + 1118
	Last calibrated:	28-29 October 2020
	Next calibration:	October 2021
	Manufacturer:	Schmid & Partner Engineering AG (CH)
E-field Probe	Type:	EX3Dv4
	Serial No:	3875
	Last calibrated:	11 June 2020
	Next calibration:	June 2021
	Manufacturer:	Schmid & Partner Engineering AG
TDS-B1 System	Type:	2x H1TDSx
	S/N Probe + RU:	1071 + 1132, 1094 + 1147
	Last calibrated:	27-29 October 2020
	Next calibration:	October 2021
	Manufacturer:	Schmid & Partner Engineering AG (CH)
Data Acquisition Systems	Type:	DAE4
	Serial No:	1553
	Last calibrated:	8 July 2020
	Next calibration:	July 2021
	Manufacturer:	Schmid & Partner Engineering AG (CH)
Thermometer	Type:	DTM 3000
	Serial No:	3020
	Last calibrated:	22 October 2020
	Next calibration:	October 2021
	Manufacturer:	LKM GmbH (DE)

Table 3: Principal measurement equipment.

Verification Source	Type:	CLA-6
	Inventory No.:	1001
	Manufacturer:	Schmid & Partner Engineering AG (CH)
	Last calibrated:	19 April 2021
	Calibration Uncertainty:	$\pm$ 18.4 % ($\varkappa = 2$)@ 6 MHz, 1 g avg.
		$\pm$ 18 % ($\varkappa = 2$)@ 6 MHz, 10 g avg.

Table 4: System Check Verification Source

3 Methods

3.1 Transfer Function Measurements

Transfer function measurements were performed via piecewise excitation in the PiX System (ZMT Zurich MedTech AG). Tissue simulating medium (TSM) with high permittivity/conductivity (TLe78c0.47, ZMT, target σ = 0.47 S/m, ε=78) was used in accordance with Clause 8 of IEC/IEEE 62209-1528-2020. The target lead was mounted such that the tip electrode was aligned with the time-domain-sensing (TDS) optically isolated E-field probe (SPEAG), while the other lead was mounted perpendicularly so as not to impact the measurement (see Figure **??**). The transfer functions were also measured with the leads mounted co-linearly on the racetrack, to conservatively estimate the effects of inter-lead coupling; the deviation was determined to be minimal, and is included in the transfer function uncertainty. The tip of the intra-cochlear lead was used as the reference point of the transfer function, as the transfer function did not show phase effects in high-permittivity TSM and the electrodes making up the array are similarly wired and can be treated as a single large scatterer. Measurements were made with 1 cm resolution with the excitor positioned 1 mm above the implant's maximum height, sweeping a path 20 mm past the implant in each direction.

Before each measurement, the parameters of the TSM were determined with the calibrated Dielectric Assessment Kit System (DAKS-12, Schmid and Partner Engineering AG, SPEAG) at the target frequency of 64 or 128 MHz. Then, the excitor operating at 64 or 128 MHz was swept along the lead using DASY (SPEAG), and the complex-valued tip response recorded at 10 mm increments. Following the measurement, the test item was removed and the incident field was measured to perform background subtraction.

3.2 Deposited Power Measurement for Transfer Function Scaling and Validation

To scale and validate the transfer function models, the test items were exposed to iso-electric E-fields in the MITS1.5/3.0 (ZMT) birdcage resonator systems, operating at linear polarization, in ELIT1.5/3.0 phantoms filled with the same high permittivity/conductivity TSM, as well as low permittivity/conductivity TSM (TLe11c0.045, ZMT, target σ = 0.045 S/m, ε=11).

For scaling measurements, test items were positioned similarly to the piX, i.e. with the target lead parallel to the incident field and the other lead perpendicular, or parallel when bounding the potential effects of lead coupling. For validation of the transfer function in high-permittivity medium, fold-back measurements were performed, such that the tangential E-field was partially reversed in direction; the resulting reductions in measured deposited power were compared to those predicted using the transfer function.

Once the test item was mounted and the dielectric properties of the TSM verified, near-field SAR probes (EX3D, SPEAG) were used with DASY (SPEAG) to measure the E-field enhancement at three planes above the target electrode, as well as the incident field after removal of the racetrack.

The measurements were co-registered to simulated distributions of the deposited power around each electrodes following the "touchless" method [4]. The deposited power distributions were simulated via a plane-wave excitation of a numerical surrogate of each lead tip, derived from the CAD models provided by Cochlear, using the verified FDTD solver of Sim4Life (ZMT). Following the co-registration, the deposited power simulations were scaled using simulations to estimate the total deposited power, which were normalized to the measured incident fields and used to scale the measured normalized transfer functions.

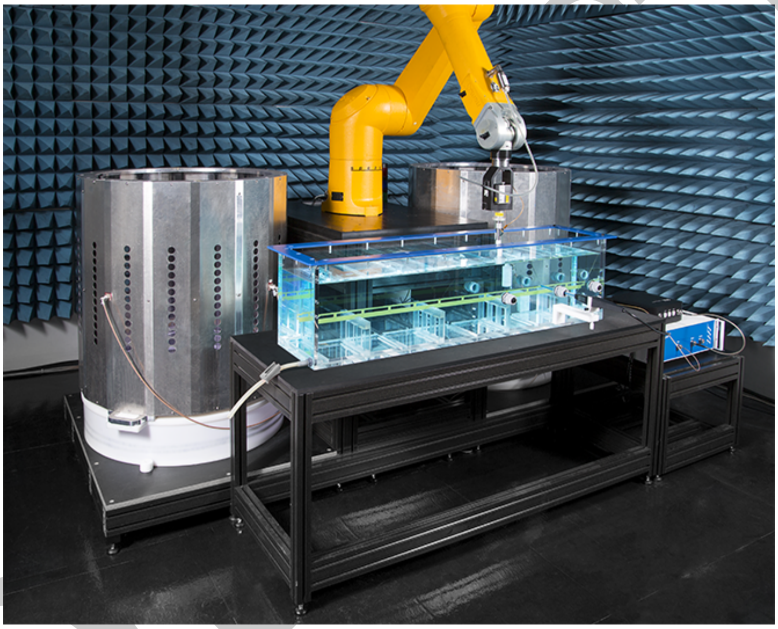

Figure 2: MITS1.5T (left) and 3.0T (right) with piX system (front) and DASY with piX excitor (back).

3.3 Simulation Tools

Software Tools	Sim4Life	6.0
	IMAnalytics	3.0
	ViP	V3.1 (Fats V3.2)
	MRIxViP	2.1

Table 5: Software tool versions used in the study.

Sim4Life

Numerical simulations were performed with the latest version of the multi-physics Sim4Life simulation platform (ZMT Zurich MedTech AG), which allows powerful physics solvers to be combined with advanced tissue models for direct characterization of real-world biological phenomena. For this project, the full-wave FDTD solver of Maxwell's equations was used, as well as the thermal solver, which is based on a modified version of the Pennes bioheat equation with advanced perfusion and thermoregulation. Sim4Life modules are subject to extensive and thorough quality assurance procedures, which include nightly test suites, unit tests, issue tracking and traceable versioning. Both FDTD and thermal solvers have been systematically verified; the verification report covers are shown in Annex A.

Virtual Population and IT'IS Tissue Database

The Virtual Population (ViP) is a set of detailed high-resolution anatomical models created from magnetic resonance image data of volunteers. Originally developed by IT'IS in collaboration with the FDA, the newest generation of ViP 3.x have more than 300 tissues and organs per model and a resolution of 0.5 x 0.5 x 0.5 mm^3 throughout the entire body. The anatomy, physiology, and segmentation/generation procedures of the ViP 3.x family have been verified and documented.

The IT'IS database of tissue properties provides material parameter values for a wide range of tissues and materials, drawn from comprehensive scientific literature review. The database also includes statistical information about the spread and standard deviation per tissue for the different parameters, which is important for assessment of the contribution to the uncertainty in a quantity of interest due to the selection of the material parameters. Cited and applied in hundreds of published studies and papers, the ViP models and the IT'IS material parameter database are continually updated and extended. Traceability is assured through version control associated with unique Digital Object Identifiers (DOI); previous versions remain accessible, together with a log file documenting the changes.

MRIxViP

IT'IS Foundation has conducted experimental surveys of commercial MRI scanners, where the strengths of the fields (B_0, B_1 and G) during pre-selected scan procedures were measured; the physical dimensions of the RF coils may be approximated from the measured B_1 profiles. Based on the experimental surveys and the product survey of vendors' specifications, the Birdcage Library (BCLib) was developed, consisting of 20 cylindrical RF birdcage body coils of different dimensions, consistent with the reference values given for commercial 1.5 T and 3 T scanners (10 of each type) in the annex of ISO/TS 10974.

The MRIxViP libraries consists of 3D distributions of electromagnetic fields that are representative for the 64 and 128 MHz RF exposure of any person undergoing any MR examination scan with a high-pass birdcage body coil. The local fields induced in the human body are a function of multiple factors:

- birdcage (port excitations, dimensions);
- anatomy (age, height, weight, body mass index, posture, etc.);
- landmark position.

In order to guarantee the appropriate population coverage, the library combines birdcage variations covering all of the MRI birdcages (BCLib), anatomical variations (ViP 3.x) and all landmark positions in 50 mm steps from head to feet. For each of the variations, the library includes the induced 3D electric and magnetic fields at a resolution finer than 2 x 2 x 2 mm for both coil excitation ports (I and Q) separately, such that any polarization can be obtained by superposition.

The combined IMAnalytics with MRIxViP1.5T/3.0T and BCLib Toolset (Figure 4) is qualified by the FDA [5] as a Medical Device Development Tool (MDDT) for use in the context of Tier 2 and Tier 3 evaluations per IEC/IEEE 62209-1528-2020.

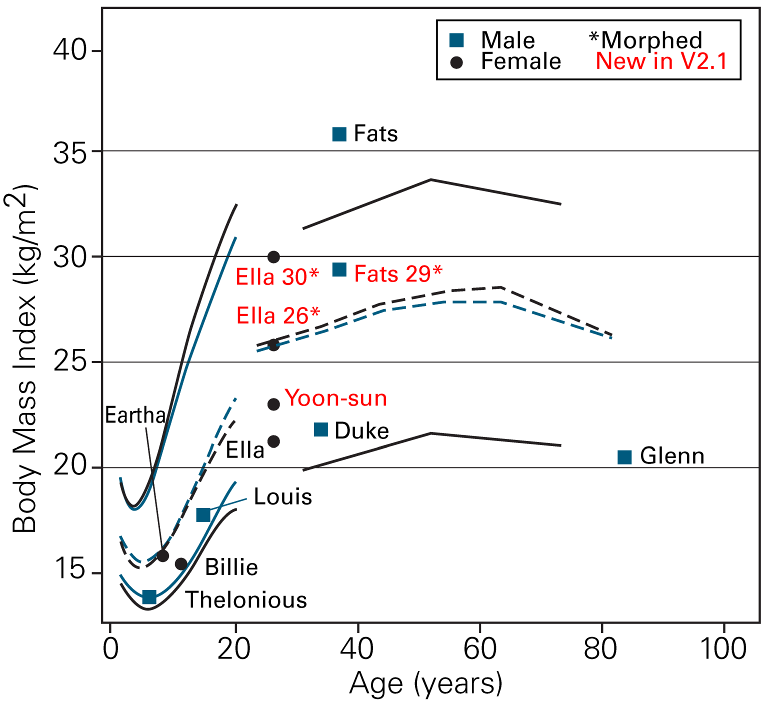

Figure 3: Population coverage of the ViP models in MRIxViP (males: blue squares; females: black circles) with age-dependent statistical data (dashed lines: 50th percentile; solid lines: 5th and 95th percentiles; sources: 2–20 years from National Center for Health Statistics (2000); 25–80 years approximated from Ogden *et al.*). Morphed ViP models of Ella and Fats are used to improve the population coverage.

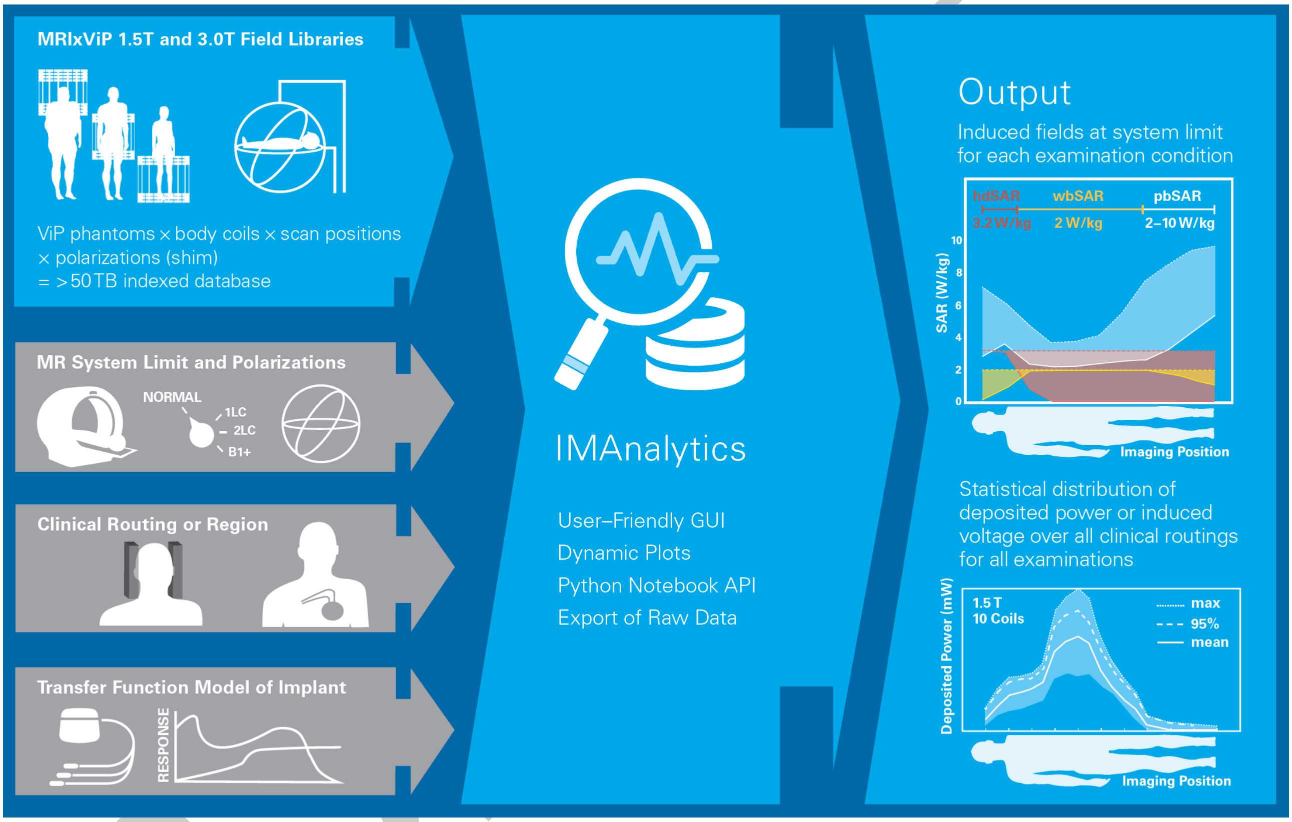

Figure 4: MDDT IMAnalytics with MRIxViP: validated, automated Tier 3 (and Tier 4-) analysis of *in vivo* response of implant transfer functions with user-defined normalizations, coil dimensions, polarizations, clinical routings, subsets of the patient model population, and landmarks) [5].

4 Equipment Under Test (EUT)

The equipment under test (EUT) was the EUTname (Figure 5), as described in Table 6, which was provided by Customer Name. The EUT was received on 7 September 2020 at the IT'IS Foundation headquarters in Zeughausstrasse 43, 8004 Zurich, and was inspected and found to be in good order. The device was evaluated as a black box, i.e., no further verification regarding the appropriate function of the device was conducted by IT'IS.

Model Type:	EUTname
Inventory No.:	1
Manufacturer:	Customer Name
Test Frequency	6.78 MHz

Table 6: Equipment under test (EUT)

Figure 5: Equipment Under Test (EUT) a wireless charging system. Shown is the power transmitter with transmitting loop (left) and the receiving implant mock-up (right); not to scale.

4.1 Equipment Under Test Output Voltage Monitoring

To monitor the output of the EUT, Customer Name suggested to measure the output voltage across the output terminal. The setup for this is shown in Figure 6. The output voltage was measured at the start of each SAR measurement. The results are used to scale the SAR result to the nominal output voltage of 514 V_{pp} provided by Customer Name.

Figure 6: Setup to monitor the output voltage of the EUT. Left: overview of the voltage monitoring setup. Right: terminal connection details of the oscilloscope probes on the EUT output.

5 Test Protocol

The test protocol employed for the SAR measurement was based on Chapter 7 of IEC/IEEE 62209-1528-2020. All tests were performed on 23-30 September 2020. Additional tests and details about the test and test conditions are outlined in the following subsections.

5.1 Tissue Equivalent Medium

For measurements in the frequency range of 6 MHz, the tissue simulating liquid “HBBL4-250V3” was used. Before the dosimetric evaluation, the parameters of the actual liquid were determined with the SPEAG Dielectric Assessment Kit (DAK3.5) (an open-ended coaxial probe). For both the test frequency of the EUT and the system check frequency, the dielectric was found to be within 10 % from the target as required by Paragraph 7.2.1b of IEC/IEEE 62209-1528-2020. The dielectric test results are summarized in Table 7.

	IEC/IEEE 62209-1528-2020		measured		deviation from target	
f/(MHz)	ε_r'	σ/(S/m)	ε_r'	σ/(S/m)	ε_r'	σ/(S/m)
5	55	0.75	52.8 ± 2.5 %	0.72 ± 2.5 %	–4.0 %	–4.0 %
10	55	0.75	52.8 ± 2.5 %	0.72 ± 2.5 %	–4.0 %	–4.0 %

Table 7: Measured dielectric parameters of the tissue simulating liquid “HBBL4-250V3” compared to the target values of IEC/IEEE 62209-1528-2020. The uncertainty of the measurement of $|\varepsilon_r|$ was estimated to be $\pm$ 2.5%. A worst-case density of 1000 kg/m^3 was used for the evaluation.

5.2 Ambient Environment

The noise level was periodically verified by conducting measurements without the EUT. The recorded ambient parameters are displayed in Table 8.

Temperature (evaluation):	22 ± 2 °C
Temperature (liquid measurement):	22 ± 2 °C
Humidity:	20–50 %
SAR Noise:	<10 µW/g

Table 8: Environmental parameters

5.3 System Check

A system check, according to clause A.2 of IEC/IEEE 62209-1528-2020, was performed before the test of the EUT to confirm the full system functionality. In this test, a calibrated system verification source (Confined Loop Antenna) fed with a 1 W continuous wave signal at 6 MHz was placed below the phantom, as shown in Figure 7. The results of the system check compared to the calibrated targets are shown in Table 9. The results confirm the system’s full functionality within the k = 1 uncertainty of the measurement system (see Table 10).

Figure 7: Setup of the system check source below the ELI4 phantom.

f /(MHz)	SAR target /(W/kg/W)		SAR measured /(W/kg/W)		deviation from target	
	1 g	10 g	1 g	10 g	1 g	10 g
6.0	0.171	0.107	0.176	0.109	+2.9 %	+1.9 %

Table 9: System check results for CLA-6.

5.4 Equipment Under Test Setup

The EUT was treated as a black box and set up according to the instructions of Customer Name During the test, the EUT was operated with the power supply that was supplied by Customer Name The transmitting frequency and the continuous wave character of the signal were verified using a spectrum analyzer over the air. Customer Name confirmed that the EUT does not apply any power control mechanisms. For the test, the transmitting loop and feeding cable where place on a RohaCell (low loss, low dielectric constant) platform (Figure 8a). This platform was used to position the loop against the test phantoms.

The test configurations were as defined below:

Flat phantom, touch: The transmitting loop was positioned in touch directly against the flat (ELI4) phantom. (Figure 8b)

Flat phantom, 2 mm: The transmitting loop was positioned with a 2 mm low loss spacer (Figure 8c) present in touch against the flat (ELI4) phantom.

Flat phantom, 5 mm: The transmitting loop was positioned with a 5 mm low loss spacer (Figure 8c) present in touch against the flat (ELI4) phantom.

Flat phantom, 10 mm: The transmitting loop was positioned with a 10 mm low loss spacer (Figure 8c) present in touch against the flat (ELI4) phantom.

Head, left, centered: The loop was centered around the ear of the left head phantom side. (Figure 8d).

Head, left, off-center: The loop was off-centered toward the back of the left head phantom side with the wire passing under the ear of the phantom (Figure 8e).

Head, right, centered: The loop was centered around the ear of the right head phantom side. (Figure 8f).

Head, right, off-center: The loop was off-centered toward the back of the right head phantom side with the wire passing under the ear of the phantom (Figure 8g).

Head, top, centered: The loop was centered on top of the head (Figure 8h). This configuration also simulates the position of the loop on the back of the head, which cannot be tested with available phantoms due to the large loop dimensions.

Implant at SAR maximum: The differential SAR change with and without the implant mock-up placed at the measured SAR maximum was tested as shown in Figure 8i.

Implant at loop center: The differential SAR change with and without the implant mock-up placed at the transmitting loop center was tested as shown in Figure 8j.

(a) Loop on RohaCell platform.

(b) EUT in touch against flat phantom.

(c) EUT with a 5 mm spacer in front of the flat phantom.

(d) EUT centered on left head.

(e) EUT off-centered on left head.

(f) EUT centered on right head.

(g) EUT off-centered on right head.

(h) EUT centered on top of head.

(i) Implant mock-up placed at the measured maximum SAR location.

(j) Implant mock-up placed at the center location (loop center).

Figure 8: Configuration of the EUT below the test phantoms.

6 Measurement Uncertainty

The measurement uncertainties are summarized in Table 10 for the head (Twin-SAM) and flat (ELI4) phantoms and Table 11 for the head-stand phantom (head, top configuration). The uncertainties are based on SAR measurement standards IEC/IEEE 62209-1528-2020.

As the EUT showed drifts larger than the 0.2 dB recommended by IEC/IEEE 62209-1528-2020, this uncertainty item has been zeroed and the test results are reported by scaling them to the worst-case drift determined.

DASY6 Uncertainty Budget
According to IEC/IEEE 62209-1528-2020
(Hand-Held/Body-Worn: 4 MHz - 3 GHz range)

Error Description	Uncert. value	Prob. Dist.	Div.	(c_i) 1g	(c_i) 10g	Std. Unc. (1g)	Std. Unc. (10g)	(v_i) v_{eff}
Measurement System								
Probe Calibration	±6.7 %	N	1	1	1	±6.7 %	±6.7 %	∞
Axial Isotropy	±4.7 %	R	$\sqrt{3}$	0	0	±0.0 %	±0.0 %	∞
Hemispherical Isotropy	±9.6 %	R	$\sqrt{3}$	1	1	±5.5 %	±5.5 %	∞
Boundary Effects	±0.0 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Linearity	±4.7 %	R	$\sqrt{3}$	1	1	±2.7 %	±2.7 %	∞
System Detection Limits	±1.0 %	R	$\sqrt{3}$	1	1	±0.6 %	±0.6 %	∞
Modulation Response[m]	±2.4 %	R	$\sqrt{3}$	1	1	±1.4 %	±1.4 %	∞
Readout Electronics	±0.3 %	N	1	1	1	±0.3 %	±0.3 %	∞
Response Time	±0.8 %	R	$\sqrt{3}$	1	1	±0.5 %	±0.5 %	∞
Integration Time	±2.6 %	R	$\sqrt{3}$	1	1	±1.5 %	±1.5 %	∞
RF Ambient Noise	±3.0 %	R	$\sqrt{3}$	1	1	±1.7 %	±1.7 %	∞
RF Ambient Reflections	±3.0 %	R	$\sqrt{3}$	1	1	±1.7 %	±1.7 %	∞
Probe Positioner	±0.02 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Probe Positioning	±2.9 %	R	$\sqrt{3}$	1	1	±1.6 %	±1.6 %	∞
Max. SAR Eval.	±15.0 %	R	$\sqrt{3}$	0	0	±0.0 %	±0.0 %	∞
Test Sample Related								
Device Positioning	±2.9 %	N	1	1	1	±2.9 %	±2.9 %	145
Device Holder	±3.6 %	N	1	1	1	±3.6 %	±3.6 %	5
Power Drift	±5.0 %	R	$\sqrt{3}$	1	1	±2.9 %	±2.9 %	∞
Power Scaling[p]	±0 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Phantom and Setup								
Phantom Uncertainty	±6.1 %	R	$\sqrt{3}$	1	1	±3.5 %	±3.5 %	∞
SAR correction	±1.9 %	N	1	1	0.84	±1.9 %	±1.6 %	∞
Liquid Conductivity (mea.)[DAK]	±2.5 %	N	1	0.78	0.71	±2.0 %	±1.8 %	∞
Liquid Permittivity (mea.) [DAK]	±2.5 %	N	1	0.23	0.26	±0.6 %	±0.7 %	∞
Temp. unc. - Conductivity [BB]	±3.4 %	R	$\sqrt{3}$	0.78	0.71	±1.5 %	±1.4 %	∞
Temp. unc. - Permittivity [BB]	±0.4 %	R	$\sqrt{3}$	0.23	0.26	±0.1 %	±0.1 %	∞
Combined Std. Uncertainty						±11.8 %	±11.7 %	1329
Expanded STD Uncertainty						**±23.7 %**	**±23.5 %**	

Table 10: Worst-Case uncertainty budget for DASY6 assessed according to IEC/IEEE 62209-1528-2020for the Twin-SAM and ELI4 phantoms. The budget is valid for the frequency range 4 MHz - 3 GHz and represents a worst-case analysis.
Footnote details: [m] SMC calibration is a new method for determining the total deviation from linearity. For psSAR of approximately 10 W/kg the uncertainty is 9.6%. For psSAR of less than 2 W/kg the uncertainty is less than 2.4% (see modulation calibration parameter uncertainty in the probe calibration certificate); [BB] if SPEAG's broad-band liquids (BBL) are used that have low temperature coefficients; [DAK] if SPEAG's high precision dielectric probe kit (DAK) is applied; [p] if power scaling is used, error item "Power Scaling" must be adjusted accordingly.

DASY6 Uncertainty Budget
According to IEC/IEEE 62209-1528-2020
(Hand-Held/Body-Worn: 4 MHz - 3 GHz range)

Error Description	Uncert. value	Prob. Dist.	Div.	(c_i) 1g	(c_i) 10g	Std. Unc. (1g)	Std. Unc. (10g)	(v_i) v_{eff}
Measurement System								
Probe Calibration	±6.7 %	N	1	1	1	±6.7 %	±6.7 %	∞
Axial Isotropy	±4.7 %	R	$\sqrt{3}$	0	0	±0.0 %	±0.0 %	∞
Hemispherical Isotropy	±9.6 %	R	$\sqrt{3}$	1	1	±5.5 %	±5.5 %	∞
Boundary Effects	±0.0 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Linearity	±4.7 %	R	$\sqrt{3}$	1	1	±2.7 %	±2.7 %	∞
System Detection Limits	±1.0 %	R	$\sqrt{3}$	1	1	±0.6 %	±0.6 %	∞
Modulation Response^{m}	±2.4 %	R	$\sqrt{3}$	1	1	±1.4 %	±1.4 %	∞
Readout Electronics	±0.3 %	N	1	1	1	±0.3 %	±0.3 %	∞
Response Time	±0.8 %	R	$\sqrt{3}$	1	1	±0.5 %	±0.5 %	∞
Integration Time	±2.6 %	R	$\sqrt{3}$	1	1	±1.5 %	±1.5 %	∞
RF Ambient Noise	±3.0 %	R	$\sqrt{3}$	1	1	±1.7 %	±1.7 %	∞
RF Ambient Reflections	±3.0 %	R	$\sqrt{3}$	1	1	±1.7 %	±1.7 %	∞
Probe Positioner	±0.02 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Probe Positioning	±2.9 %	R	$\sqrt{3}$	1	1	±1.6 %	±1.6 %	∞
Max. SAR Eval.	±15.0 %	R	$\sqrt{3}$	1	1	±8.7 %	±8.7 %	∞
Test Sample Related								
Device Positioning	±2.9 %	N	1	1	1	±2.9 %	±2.9 %	145
Device Holder	±3.6 %	N	1	1	1	±3.6 %	±3.6 %	5
Power Drift	±5.0 %	R	$\sqrt{3}$	1	1	±2.9 %	±2.9 %	∞
Power Scaling^{p}	±0 %	R	$\sqrt{3}$	1	1	±0.0 %	±0.0 %	∞
Phantom and Setup								
Phantom Uncertainty	±6.1 %	R	$\sqrt{3}$	1	1	±3.5 %	±3.5 %	∞
SAR correction	±1.9 %	N	1	1	0.84	±1.9 %	±1.6 %	∞
Liquid Conductivity (mea.)^{DAK}	±2.5 %	N	1	0.78	0.71	±2.0 %	±1.8 %	∞
Liquid Permittivity (mea.) ^{DAK}	±2.5 %	N	1	0.23	0.26	±0.6 %	±0.7 %	∞
Temp. unc. - Conductivity ^{BB}	±3.4 %	R	$\sqrt{3}$	0.78	0.71	±1.5 %	±1.4 %	∞
Temp. unc. - Permittivity ^{BB}	±0.4 %	R	$\sqrt{3}$	0.23	0.26	±0.1 %	±0.1 %	∞
Combined Std. Uncertainty						±14.7 %	±14.6 %	1329
Expanded STD Uncertainty						**±29.3 %**	**±29.2 %**	

Table 11: Worst-Case uncertainty budget for DASY6 assessed according to IEC/IEEE 62209-1528-2020for the head stand phantom. The budget is valid for the frequency range 4 MHz - 6 GHz and represents a worst-case analysis.
Footnote details: ^{m} SMC calibration is a new method for determining the total deviation from linearity. For psSAR of approximately 10 W/kg the uncertainty is 9.6%. For psSAR of less than 2 W/kg the uncertainty is less than 2.4% (see modulation calibration parameter uncertainty in the probe calibration certificate); ^{BB} if SPEAG's broad-band liquids (BBL) are used that have low temperature coefficients; ^{DAK} if SPEAG's high precision dielectric probe kit (DAK) is applied; ^{p} if power scaling is used, error item "Power Scaling" must be adjusted accordingly.

7 Confidence Interval

The confidence intervals are estimated as described in CR-0540/0741 and extensions, in accordance with the methodology of the GUM and ISO/TR 21900.

7.1 Validation Sample Confidence Interval

The uncertainty associated with physical variation of validation samples, provided by Customer Name with a specific wire broken at a specific location, is estimated by sample-to-sample variation of the measured scaled transfer function of electrode 1 of three samples with wire 1 broken at 29.5 cm from the tip, each from a different manufacturing batch. The variation in of isoelectric deposited power is used to estimate the confidence interval of the scaling factor, and difference in Tier 3 95% deposited power in Duke for the same scaling factor is used to quantify deviation of the relative transfer function.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**
Scaling factor	0.05	N	1	1	0.05
Transfer Function	0.32	N	1	1	0.32
Combined Std. Uncertainty (k = 1)					**0.32**

Table 12: Uncertainty from physical variation of validation samples.

7.2 Measurement Confidence Interval

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**
Analyzer drift	0.01	N	2	1	0.01
Cable movement	0.15	N	2	1	0.08
TDS linearity	0.50	N	2	1	0.25
Excitor positioning (x,y)	0.10	N	2	1	0.05
Excitor positioning (z)	0.50	N	2	1	0.25
Phantom boundaries	0.40	N	2	1	0.20
Reflected power	0.12	N	2	1	0.06
TSM properties	0.25	N	2	1	0.13
Device definition	0.10	N	2	1	0.05
Post-processing	0.14	N	2	1	0.07
Combined Std. Uncertainty (k = 1)					**0.45**

Table 13: Uncertainty of the normalized transfer function model of the test item.

Source of Uncertainty	Tolerance in dB	Prob. Distrib.	Div.	c_i	Std. Unc. in dB
$B_{1,RMS}$	–	–	–	0	0
$B_{1,RMS}$ drift	0.20	R	$\sqrt{3}$	1	0.12
Phantom position	0.71	R	$\sqrt{3}$	1	0.41
Phantom fill-height	0.20	R	$\sqrt{3}$	1	0.12
Test item position	1.00	R	$\sqrt{3}$	1	0.58
TSM conductivity	0.30	R	$\sqrt{3}$	1	0.17
TSM permittivity	0.25	R	$\sqrt{3}$	1	0.15
Polarization	0.27	R	$\sqrt{3}$	1	0.16
TFD incident field	1.17	R	$\sqrt{3}$	1	0.68
Combined Std. Uncertainty (k = 1)					**0.92**

Table 14: Combined uncertainty of incident field exposure, due to the RF coil and phantom.

Source of Uncertainty	Tolerance in dB	Prob. Distrib.	Div.	c_i	Std. Unc. in dB
DASY SAR Enhancement					
Basic calibration	0.42	N	2	0	0
Modulation	0.01	R	$\sqrt{3}$	0	0
TSM parameter	0.54	N	2	0	0
Frequency response	0.27	N	2	0	0
Axial isotropy	0.15	R	$\sqrt{3}$	0	0
Probe linearity	0.20	R	$\sqrt{3}$	1	0.12
Spherical isotropy	0.60	R	$\sqrt{3}$	1	0.35
Readout electronics	0.01	R	$\sqrt{3}$	1	0.01
Probe holder	0	R	$\sqrt{3}$	1	0
Integration time	0	R	$\sqrt{3}$	1	0
Probe positioning	–	–	–	–	–
Interpolation, averaging	–	–	–	–	–
"Touchless" Co-Registration					
Co-registration	0.90	R	$\sqrt{3}$	1	0.52
Background inhomogeneity	0.50	R	$\sqrt{3}$	1	0.29
AIMD holder	0.60	R	$\sqrt{3}$	1	0.35
Volume integral ($\leq$–40 dB)	0.50	R	$\sqrt{3}$	1	0.29
Grid resolution	0.25	R	$\sqrt{3}$	1	0.14
Boundary condition ($\leq$–80 dB)	0.01	R	$\sqrt{3}$	1	0.01
Tip modeling	0.51	R	$\sqrt{3}$	1	0.29
Combined Std. Uncertainty (k = 1)					**0.89**

Table 15: Uncertainty of local SAR enhancement measured via radiated testing in the MITS using DASY with an EX3D SAR probe.

7.3 Model and Simulation Confidence Intervals

The uncertainty of the numerical derivation of the transfer function is estimated by differences in predictions between the two approaches for the unbroken case. The variation in of isoelectric deposited power is used to estimate the confidence interval of the scaling factor, and difference in Tier 3 95% deposited power in Duke for the same scaling factor is used to quantify deviation of the relative transfer function. The influence of the IPG model is estimated by the maximum variation in isoelectric deposited power for each unbroken case for two models with different ground planes.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**
Scaling factor	1.0	N	1	1	1.0
Transfer Function	0.8	N	1	1	0.8
IPG model	1.9	N	1	1	1.9
Combined Std. Uncertainty (k = 1)					**2.29**

Table 16: Uncertainty of numerical device model and transfer function approach used for test reduction.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**
Absorbing boundary condition	0.04	N	1	1	0.04
Simulation time	0.20	R	$\sqrt{3}$	1	0.12
Grid resolution	0.17	N	1	1	0.17
Insulation radius	0.10	N	1	1	0.10
Insulation dielectric constant	0.80	R	$\sqrt{3}$	1	0.46
Combined Std. Uncertainty (k = 1)					**0.52**

Table 17: Uncertainty of numerical modeling of the 3D SAR distribution of the electrodes *in vitro*.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**
Transfer function in complex tissues	1.0	N	1	1	1.00
MRIxViP fields modeling	0.2	N	1	1	0.2
Population anatomy	–	–	–	–	–
RF-coil	–	–	–	–	–
Clinical routing	0.15	R	$\sqrt{3}$	1	0.09
ViP model and tissue parameters	1.37	R	$\sqrt{3}$	1	0.79
Patient position	0.5	N	1	1	0.5
B_1 exposure limit	0.1	R	$\sqrt{3}$	1	0.06
Combined Std. Uncertainty (k = 1)					**1.39**

Table 18: Standard uncertainty of IMAnalytics output of deposited power due to uncertainty of *in vivo* exposure conditions (MRIxViP library) and influence of heterogeneous tissues on transfer function. [5]

7.4 Combined Confidence Interval

The combined uncertainty of the measured scaled transfer function of the sample with 1 wire break at 29.5 cm, comprising sample uncertainty and measurement uncertainty, is given in Table 19. The combined uncertainty of the Tier 3 evaluation made with this transfer function is given in Table 20. The uncertainty of the Tier 3 results based on the numerical transfer functions, used for test reduction (identification of worst-case) is given in Table 21.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**	**Ref.**
Sample Variation	0.32	N	1	1	0.32	Table 12
Normalized Transfer Function	0.45	N	1	1	0.45	Table 13
Incident Field Exposure	0.92	N	1	1	0.92	Table 14
Deposited Power Measurement	0.89	N	1	1	0.89	Table 15
Combined Std. Uncertainty (k = 1)					**1.4**	

Table 19: Combined uncertainty of the measured scaled transfer function of the LEAD1058-50B sample with 1 wire break at 29.5 cm.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**	**Ref.**
Measured Transfer Function	1.39	N	1	1	1.39	Table 19
Tier 3 Evaluation	1.39	N	1	1	1.39	Table 18
Combined Std. Uncertainty (k = 1)					**2.0**	

Table 20: Combined uncertainty of the Tier 3 evaluation made with the measured scaled transfer function of sample at 29.5 cm.

Source of Uncertainty	**Tolerance in dB**	**Prob. Distrib.**	**Div.**	c_i	**Std. Unc. in dB**	**Ref.**
Device Model and Transfer Function	2.29	N	1	1	2.29	Table 16
Numerical Model of SAR Distribution	0.52	N	1	1	0.52	Table 17
Tier 3 Evaluation	1.39	N	1	1	1.39	Table 18
Combined Std. Uncertainty (k = 1)					**2.7**	

Table 21: Uncertainty of numerical device model and transfer function approach used for test reduction.

8 Test Results

The following sections summarize the test results. Section 8.1 presents the terminal voltages measured at the beginning of each test. Section 8.2 provides the test results in terms of SAR over all test positions. Section 8.3 provides the relative change in local spatially averaged SAR when an implant is present.

8.1 Terminal Voltages

Table 22 summarizes the voltages measured at the start of each SAR test configuration measurement.

Test Configuration	**Voltage /V_{pp}**
Flat phantom, touch	394
Flat phantom, 2 mm	397
Flat phantom, 5 mm	397
Flat phantom, 10 mm	405
Head, left, centered	440
Head, left, off-center	440
Head, right, centered	445
Head, right, off-center	436
Head, top, centered	440
Implant at SAR maximum	400
No implant at SAR maximum (reference)	400
Implant at loop center	395
No implant at loop center (reference)	400

Table 22: SAR test result of the EUT.

8.2 Absolute SAR Results

The following sections document the measured SAR results (Section **??**) and as the EUT showed large deviations from the nominal terminal voltages and large drifts, SAR results scaled by voltage difference (Section **??**) as well as drift (Section **??**).

The SAR values as measured with the DASY6 system are summarized in Table 23. Also included in Table 23 is the SAR drift for all tested configurations. Drift is determined by the difference between a reference location measurement at the beginning and at the end of each measurement. The worst-case drift was 1.44 dB. During a warm up period the EUT was drifting additionally, from 500 V_{pp} to approximately 400 – 450 V_{pp}. SAR measurements were only started after a waiting period of five minutes.

For the head-left and head-right centered configurations the SAR maximum was found in the nose region. Due to mechanical restrictions the 10 g volume SAR could not be determined in these locations.

Volume SAR distributions and locations of the 1 g and 10 g SAR maxima are shown in Figures 9, 10, and 11.

Test configuration	1 g psa. SAR /(W/kg)	10 g psa. SAR /(W/kg)	drift /dB
Flat phantom, touch	1.6 ± 23.7% *	1.07 ± 23.5% *	-0.87
Flat phantom, 2 mm	1.41 ± 23.7% *	0.973 ± 23.5% *	-0.21
Flat phantom, 5 mm	1.14 ± 23.7% *	0.818 ± 23.5% *	-1.29
Flat phantom, 10 mm	0.901 ± 23.7% *	0.668 ± 23.5% *	-0.18
Head, left, centered	0.852 ± 23.7% *	n.a.	0.93
Head, left, off-center	0.995 ± 23.7% *	0.623 ± 23.5% *	1.44
Head, right, centered	1.12 ± 23.7% *	0.695 ± 23.5% *	-0.04
Head, right, off-center	1.29 ± 23.7% *	n.a.	1.19
Head, top, centered	0.462 ± 29.3%*	0.374 ± 29.2%*	0.13

* Worst-case uncertainty of the DASY6 system with specialized phantoms ($k = 2$)

Table 23: Measured SAR test result of the EUT.

(a) 0 mm distance; 0 dB = 2.5 W/kg

(b) 2 mm distance; 0 dB = 2.5 W/kg

(c) 5 mm distance; 0 dB = 2.5 W/kg

(d) 10 mm distance; 0 dB = 2.5 W/kg

Figure 9: Volume SAR distributions of the EUT measured in the flat (ELI4) phantom.

(a) Left head side, loop centered; 0 dB = 2.5 W/kg

(b) Left head side, loop off-center; 0 dB = 2.5 W/kg

(c) Right head side, loop centered; 0 dB = 2.5 W/kg

(d) Right head side, loop off-center; 0 dB = 2.5 W/kg

Figure 10: Volume SAR distributions of the EUT measured in the head (Twin-SAM) phantom.

(a) Top view; 0 dB = 2.5 W/kg

(b) Rear view; 0 dB = 2.5 W/kg

(c) Left view; 0 dB = 2.5 W/kg

(d) Right view; 0 dB = 2.5 W/kg

Figure 11: Volume SAR distribution of the EUT measured on top of the head (Head-Stand) phantom.

8.3 SAR Results with and without Implant

To assess the impact of the implant on the SAR, a fix volume above the implant (approximately 5 mm above) was measured with and without the implant present. This test was performed in the flat phantom touch configuration at the previously determined maximum SAR location, as well as at the phantom/loop center. For each test, the 1 g and 10 g SAR was determined inside the fixed test volume. When positioned in the center, the implant does not have an effect on the averaged SAR results (Figure 12). When put at the SAR maximum, i.e., directly above the loop wire, the implant can locally shield the field and can cause an enhancement at the boundaries of the can (Figure 13). This is also confirmed by a stronger increase of the 1 g SAR value of 0.48 dB, in comparison to an increase of 0.22 dB of the 10 g value.

Test configuration	Location	Implant	1 g psa. SAR /(W/kg)	10 g psa. SAR /(W/kg)
Flat phantom, touch	SAR maximum	not present	0.776	0.598
Flat phantom, touch	SAR maximum	present	0.867	0.63
Flat phantom, touch	center	not present	0.262	0.175
Flat phantom, touch	center	present	0.257	0.171

Table 24: SAR test result of the EUT.

(a) With implant 0 dB = 0.4 W/kg

(b) Without implant 0 dB = 0.4 W/kg

Figure 12: Volume SAR distribution of the EUT measured with and without implant present in the center of the loop/phantom.

(a) With implant 0 dB = 1.5 W/kg

(b) Without implant 0 dB = 1.5 W/kg

Figure 13: Volume SAR distribution of the EUT measured with and without implant present at the maximum SAR location of the loop/phantom. The bottom SAR plane in both plots was measured without the implant present to determine maximum SAR location.

9 Tier 3 Results

Approach	Scaling Factor	Transfer Function
Reciprocal Simulation	1.9	1.8
Plane-Wave Simulation	1.8	2.1

Table 25: Deviation, in dB, of numerical models for the undamaged 50 cm lead from the measured value (CR-0540/0741). Deviation in isoelectric deposited power quantifies the difference in scaling factor, while difference in Tier 3 95% deposited power in Duke for the same scaling factor is used to quantify deviation of the relative transfer function.

ViP Phantom	95% Deposited Power at Max. Landmark (mW)
Duke	327
Ella	154
Fats	205
Billie	180
Thelonious	146

Table 26: Tier 3 95% deposited power of the measured lead with wire 1 broken at 29.5 cm from the tip at the maximum landmark, evaluated over all RF body coils and clinical routings defined in projects CR-0540/0741.

Figure 14: Tier 3 95% deposited power for Duke, Ella, Fats, Billie and Thelonious, of the lead with wire 1 broken at 29.5 cm from the tip, evaluated over all RF body coils and clinical routings defined in projects CR-0540/0741.

10 Conclusions

Should be similar but different from executive summary. May also mention subsequent work.

Front Page Authors, Zurich, DD Month 2021

References

[1] International Commission on Non-Ionizing Radiation Protection (ICNIRP), "Guidelines for limiting exposure to time-varying electric, magnetic, and electromagnetic fields (up to 300 GHz)", *Health Physics*, vol. 74, no. 4, pp. 494–522, Apr. 1998.

[2] European Commission (EC), "EC1999/519/ec/1999, on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz)", Tech. Rep., European Commission, 1999.

[3] "US Code for Federal Regulations Title 47 Chapter I Subchapter A Part 1 Subpart I § 1.1310 radiofrequency radiation exposure limits.", May 2021.

[4] Aiping Yao, Earl Zastrow, and Niels Kuster, "Data-driven experimental evaluation method for the safety assessment of implants with respect to RF-induced heating during MRI", *Radio Science*, vol. 53, no. 6, pp. 700–709, 2018.

[5] US Food and Drug Administration (FDA), "MDDT summary of evidence and basis of qualification decision for IMAnalytics with MRIxViP1.5T/3.0T and BCLIB", Tech. Rep., 2021.

A Cover Pages of Verification Reports

B About Us

Z43 (www.z43.swiss)

Zurich43 (Z43) is a strategic alliance composed of four partner organizations: the nonprofit Foundation for Research on Information Technologies in Society (IT'IS) and three commercial SMEs – Schmid and Partner Engineering AG (SPEAG), ZMT Zurich MedTech AG (ZMT) and TI Solutions AG (TI Solutions). Z43's dedicated mission is to expand the boundaries of (i) accurate evaluation of electromagnetic (EM) near- and far-fields from static to optical frequencies and (ii) predictive modeling in validated anatomical and physiological environments for precision medicine. Z43 is a leading global player that collaborates with over 100 R&D centers and serves more than 500 customers worldwide. In addition, Z43 maintains two ISO17025 accredited laboratories to better serve its research partners and customers, i.e., the SPEAG Calibration Laboratory and the IT'IS Test Laboratory.

IT'IS – Foundation for Research on Information Technologies in Society (www.itis.swiss)

The IT'IS Foundation was established in 1999 through the initiative and with the support of the Federal Institute of Technology (ETH) Zurich and the global wireless communications industry, together with several governmental agencies. IT'IS is the leading independent nonprofit research institute dedicated to improving the quality of people's lives by advancing personalized medicine and computational life sciences and beneficial applications of EM energy and wireless communications (EM Research). IT'IS supports the R&D efforts of the alliance members, as well as its many academic and industrial partners, to advance precompetitive and non-competitive research initiatives; and offers a variety of customized research solutions to the wireless and medical device industries, to academic and national institutions, and to governments and regulatory bodies. IT'IS is very active in numerous standards, including IEC 62209, 62232, 62704, 63195, and 63184 (wireless transmitting devices), and IEC 60601-2-33, ISO 10974, and ASTM F2182 (magnetic resonance systems and implant safety).

SPEAG – Schmid & Partner Engineering AG (www.speag.swiss)

SPEAG was founded in 1994 to develop and manufacture EM systems and components as a spin-off of the ETH Zurich's Bioelectromagnetics/EM Compatibility research group (which later became the IT'IS Foundation). SPEAG is the leading developer and manufacturer of advanced, efficient, and reliable test equipment for the evaluation of the EM near- and far-fields at frequencies from a few kHz up to 110 GHz. SPEAG's key products are: DASY8 – specific absorption rate (SAR) measurements for safety compliance; cSAR3D – fast SAR testing; ICEy – automated near-field scanning for EM interference and compatibility (EMI/EMC); MAGPy – exposure assessments below 10 MHz; DAK – dielectric measurement systems; EM Phantoms – body simulators for radiofrequency (RF) testing; and SEMCAD X – RF performance modeling of devices used in and on the human body.

ZMT – ZMT Zurich MedTech AG (www.zmt.swiss)

ZMT was founded in 2006 as a spin-off company of ETH Zurich and the IT'IS Foundation, with the mission to develop innovative and validated simulation tools and best practices for the analysis and prediction of complex, multifaceted, and dynamic biological processes and interactions. ZMT's flagship product is Sim4Life, a state-of-the-art simulation platform that combines computable human phantoms with powerful physics solvers and the most advanced tissue models. Sim4Life is used to analyze real-world biological phenomena and complex technical medical devices and therapies in validated computational biological and anatomical environments. ZMT also provides fully characterized and ISO17025-calibrated measurement systems for model generation, verification, and validation of *in silico*-based evaluations.

TI Solutions – TI Solutions AG (www.temporalinterference.com)

TI Solutions is a young start-up developing highly flexible stimulation devices and planning tools to support investigations of non-invasive temporal interference stimulation of brain and peripheral nervous system activity. The long-term goal is to enable personalized treatments by providing the most advanced stimulation devices.

IT'IS Test Laboratory (www.itis.swiss/customized-research/)

The ISO 17025 accredited IT'IS Test Laboratory performs accurate electromagnetic near-field evaluations. Its scope is to (i) design, validate and perform electromagnetic and temperature measurements in free space and in lossy media, (ii) characterize test items and materials relating to such measurements, and (iii) provide service assistance in the areas of testing and verification. The primary applications are in areas of wireless and medical devices in which product standards have not been established yet. ISO 17025 calibrated probes, verified test equipment, and validated methodologies are utilized to ensure quality and traceability of tests. The scope of the ISO/IEC 17025 accreditation (Type C), multilaterally recognized by EA, IFA, and ILAC, includes:

- radiofrequency safety of medical implants in patients undergoing magnetic resonance imaging per ISO 10974 and ASTM F2182;
- specific absorption rate (SAR), induced electric (E-)field, and power density of wireless devices per all relevant standards;
- incident E- and magnetic (H-)fields of wireless devices from 3 kHz to 110 GHz per all relevant standards;
- dielectric characterization of materials.

SPEAG Calibration Laboratory (https://speag.swiss/services/cal-lab/accreditation/)

To better serve Z43 partners and customers, SPEAG established a calibration Laboratory in 2001 that is certified by the Swiss Accreditation Service for ISO/IEC 17025 accreditation and multilaterally recognized by EA, IFA, and ILAC. The laboratory provides extensive calibration services to the entire Z43 family for systems, probes, antennas, dielectric probe kits, phantoms, materials, etc. A number of satellite facilities have been co-founded, such as the SPEAG Calibration Laboratory Korea, established in 2011 in collaboration with DYMSTEC, in order to bring calibration services closer to SPEAG's global customer base.